

Title:

Telecom Customer Churn Analysis

Data Analytics Project Report

Subtitle:

Python | MySQL | Power BI

Report Structure:

1. Project Overview

This project analyses 7,043 telecom customers from California to identify churn drivers, revenue impact, and high-risk customer segments.

Dataset Source: Maven Analytics

Tools Used: Python, MySQL, Power BI, DAX

2. Dataset Description

Table 1: telecom_customer_churn

→ Rows: 7,043

→ Columns: 38

→ Contains: Customer demographics, services, charges, and churn information

Table 2: telecom_zipcode_population

→ Rows: 1,671

→ Columns: 2

→ Contains: ZIP code and population data

3. Data Cleaning Steps

Handled 15 columns with missing values
Filled internet service columns with

"No Internet Service"

Filled phone service columns with

"No Phone Service"

Filled offer column with "No Offer"

Zero duplicates found

Negative monthly charges identified

(120 records) — kept as valid discounts

4. Feature Engineering

7 New Features Created:

→ Age Group (6 buckets)

→ Tenure Group (6 buckets)

→ Revenue per Month

→ Is Churned (Binary)

→ Has Internet (Binary)

→ Has Phone (Binary)

→ Number of Services

5. Key Findings

26.5% Churn Rate

1,869 out of 7,043 customers churned

\$3.68M Revenue Lost

Churned customers paid highest avg \$73/month

Month-to-Month = 46% Churn

15x higher than Two Year contracts

Fiber Optic Highest Churn

Premium users leaving despite high charges

First 12 Months Critical

New customers most vulnerable to churn

No Offer = Highest Churn

Targeted offers reduce churn significantly

6. SQL Analysis

20 Business Queries written across 3 levels:

Basic (Q1-Q5)

- Overall churn rate
- Customer status breakdown
- Top cities analysis

Intermediate (Q6-Q13)

- Age group analysis
- Top churn reasons
- Revenue analysis

Advanced (Q14-Q20)

- JOIN queries with population data
 - Customer Lifetime Value (CLV)
 - Window functions & running totals
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7. Power BI Dashboard

3 Interactive Pages:

Page 1 — Executive Summary

- KPI Cards, Donut Chart, Bar Charts
- Tree map, Line Chart, Slicers

Page 2 — Customer Deep Dive

→ Funnel Chart, Area Chart

→ Ribbon Chart, 100% Stacked Bar

Page 3 — Churn Risk & Revenue Intelligence

→ Waterfall Chart, Map Visual

→ Scatter Chart, Matrix Table

8. Business Recommendations

High Priority:

→ Convert Month-to-Month → Annual contracts

→ Launch onboarding program (0-12 months)

Medium Priority:

→ Review Fiber Optic pricing strategy

→ Targeted retention offers for
no-offer customers

9. Conclusion

This project successfully identified key churn drivers in the telecom industry. The analysis revealed that contract type, tenure, and service usage are the strongest predictors of churn.

By implementing the recommended strategies, the company can potentially reduce churn rate and recover significant lost revenue.

10. Tools & Technologies

Python 3.x — Data Cleaning & EDA

Pandas — Data Manipulation

Seaborn/Plotly — Visualization

MySQL — Data Storage & Queries

Power BI — Dashboard

DAX — Business Measures

Figure 2: Customer Deep Dive Dashboard



Figure 1: Executive Summary Dashboard

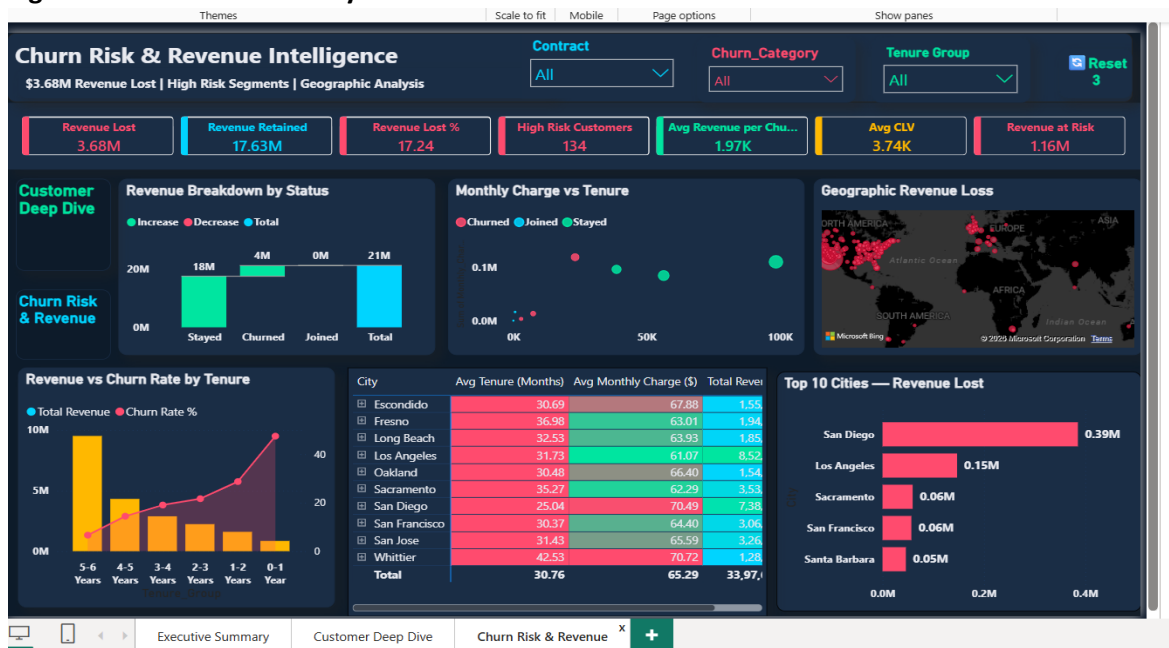


Figure 3: Churn Risk & Revenue Dashboard

